



Society of Petroleum Engineers

SPE-227029-MS

CO₂ Mineralization in Produced Water: Transforming Waste Brines into a Carbon Sink

Salem Alshammari, Hussain Saleem, Dong Kyu Cha, and Subhash Ayirala, Saudi Aramco, PE&D, EXPEC ARC Reservoir Engineering Technology, Dhahran, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227029-MS](https://doi.org/10.2118/227029-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Produced brines from oil and gas fields are rich in divalent cations, offering the potential for CO₂ sequestration as thermodynamically stable carbonates. This study investigates CO₂ mineralization in produced water through yard testing the brine alkalization process in a continuously stirred tank reactor (CSTR) having a capacity of 600 L to demonstrate the kinetics and thermodynamics of magnesium and calcium hydroxide mineral precipitation at different pH regimes. Analytical characterization of the brine chemistry was done using ion chromatography (IC), inductively coupled plasma-atomic absorption spectroscopy (ICP-AES) and pH measurements. The results from the yard test demonstrate the selective removal of magnesium hydroxide after the initial alkalization step with a pH value between 8 to 9 followed by calcium hydroxide formation after the complete removal of magnesium at a pH above 10. A holistic understanding of the effect of process parameters like temperature, pressure, solid-to-liquid ratio, and the aqueous medium ionic strength is crucial for the rational design of large-scale ex-situ CO₂ mineralization processes. Therefore, we investigated various parameters affecting the carbonation of the extracted calcium hydroxide minerals from the alkalization process using chemical thermodynamic modeling. The models were based on Pitzer's equations to account for the deviation from ideality in the electrolyte solution and compute the activities of its components whereas the gas phase fugacity was modeled using the Peng-Robinson (PR) equation of state. Increasing the temperature and reducing the pressure enhanced the CO₂ mineralization thermodynamic yield. We also studied the stepwise addition of CO₂ in a system of calcium hydroxide in water until a certain fugacity is reached (e.g., 1 atm). This simulates scenarios where CO₂ mass transfer or the hydration/hydroxylation of CO₂ is limiting. The amount of precipitated CaCO₃ increases with adding CO₂ at a stable pH, which can be between 11 and 13 depending on the temperature then CaCO₃ partially dissolves as the pH drops, so the process should be terminated immediately after the pH starts plummeting to maximize yield. This work represents an advancement in CO₂ mineralization in produced water, transitioning from laboratory-scale research to yard-scale testing using real brines. As no commercial CO₂ mineralization technology currently exists, these findings provide valuable insights for designing large-scale CO₂ mineralization processes in brine that can overcome operational challenges such as process rate, variability in feed brine chemistry, and techno-economic feasibility.

Introduction

Since the Industrial Revolution, the escalation of anthropogenic CO₂ emissions has undeniably driven global warming with the average surface temperature increasing by 1.1 °C during 2011–2020 relative to the latter half of the nineteenth century (IPCC 2023). Carbon emissions have other environmental implications besides global warming such as the absorption of CO₂ into oceans which caused the acidification of surface layers leading to a 0.02 drop in pH every ten years over the past 3 decades (Doney et al. 2009). Therefore, it is essential to develop carbon capture, utilization, and storage (CCUS) solutions to mitigate the environmental impact of CO₂ emissions. Subsurface storage is a scalable approach for carbon abatement in which CO₂ is injected and trapped in deep geological formations (Bashir et al. 2024). Nevertheless, subsurface gas storage demands substantial investment to develop the necessary infrastructure and carry out large-scale projects. This includes drilling new wells, specialized well completion to preserve well integrity with injection of corrosive CO₂ (Bai et al. 2015; Bai et al. 2016), and extensive reservoir management with surveillance and monitoring of CO₂ plumes in order to mitigate potential leakages (Ma et al. 2016). Furthermore, the complicated multiphase and interfacial interactions between CO₂, brine and rock makes predicting the fate and trapping mechanism of CO₂ difficult during geological storage (Alshammari et al. 2024a). Moreover, except for CO₂ enhanced oil recovery, geological sequestration is a one-way process involving the injection of CO₂ without generating direct revenue. Its economic viability is primarily driven by legislation and the implementation of a carbon tax (Suleman Kali et al. 2022). Ex-situ CO₂ mineralization is an alternative decarbonization scheme that can valorize carbon sequestration by producing carbonate minerals (Kelly et al. 2011). Unlike in-situ mineralization, where CO₂ is sequestered in subsurface igneous basalt or ultramafic formations (Pogge von Strandmann et al. 2019), ex-situ mineralization is a surface process that utilizes alkaline solids or divalent-rich brines to fix CO₂ as a carbonate mineral (Zhang et al. 2020). The raw materials for ex-situ mineralization can include natural minerals such as wollastonite (Wu et al. 2001; Zevenhoven et al. 2010), serpentine (Park et al. 2003), and olivine (Eikeland et al. 2015); waste solids like steelmaking slag (Santos et al. 2013), cement kiln dust (Huntzinger et al. 2018), and coal fly ash (Montes-Hernandez et al. 2009); or divalent-rich brines such as desalination rejects (Wang et al. 2011; Bang et al. 2019; Bang et al. 2021; Chen et al. 2021; Lee et al. 2024) and produced waters from oil and gas reservoirs (Druckenmiller and Maroto-Valer 2005; Soong et al. 2004; Zhu et al. 2022; Alyousef et al. 2025). Despite requiring additional dissolution step, the mineral carbonation approach using reactive minerals or solid waste has comparatively more studies in the literature compared to the brine approach (Zhang et al. 2020). Furthermore, even within the CO₂ mineralization using brines category, few papers investigated the potential of using produced water from oil and gas wells despite having a much higher concentrations of calcium ions compared to seawater desalination reject (Alyousef et al. 2025). Aquifers in deep geological formations such as in the case of petroleum systems tend to be hypersaline and very rich in calcium ions with an average concentration between 2,530 mg/L to 25,800 mg/L and can reach up to 205,000 mg/L in some Pennsylvanian systems compared to seawater which only have a calcium concentration around 400 mg/L (Collins 1975). Therefore, produced water from oil and gas fields can be a promising feed to mineralize CO₂. Nevertheless, few papers (Druckenmiller and Maroto-Valer 2005; Soong et al. 2004; Soong et al. 2006; Zhu et al. 2022; Alyousef et al. 2025) have considered produced water or subsurface brines as a feed to mineralize CO₂. Moreover, produced water can be a valuable source for the extraction of critical minerals like lithium (Alshammari et al. 2024b). In this work, we present a novel approach for effective mineral carbonation using produced water, implemented through a multi-stage process (Figure 1) that leverages the distinct thermodynamic properties of magnesium and calcium. Specifically, the process exploits the preferential alkalization of magnesium prior to calcium. The sequential alkalization was demonstrated in a yard-scale test using a continuously stirred tank reactor (CSTR) with a capacity of 600 L. Subsequently, we investigated the optimal conditions for the carbonation of the extracted calcium hydroxide, systematically evaluating the effects of pressure, temperature, and solid-to-liquid ratios to identify the most favorable parameters.

Beyond carbon sequestration and the extraction of valuable minerals such as brucite and calcium hydroxide, this process also serves to soften the brines by removing magnesium and calcium ions, thereby reducing the risk of formation damage due to inorganic scale deposition. By exploiting alkalization chemistry, this approach is adaptable to a wide range of brine compositions, with the separation and alkali dosing efficiently controlled by monitoring pH to track the progression of magnesium and calcium alkalization.

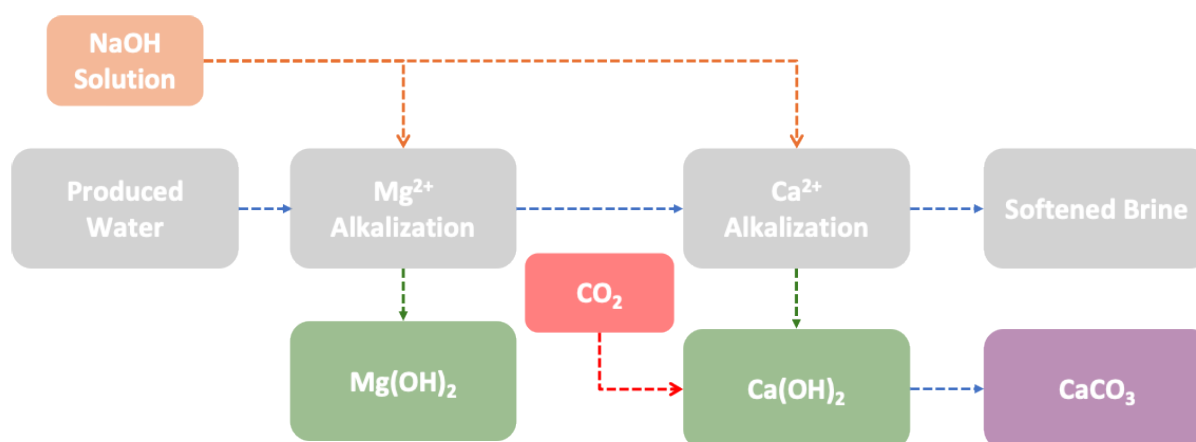


Figure 1—Mineral carbonation approach using calcium rich produced water as a feed stream.

Methods

Brine Alkalization Yard Test

A yard-scale test was conducted to investigate the kinetics and thermodynamics of brine alkalization in an upscaled system using a CSTR with a working capacity of 600 L. The reactor was equipped with an overhead mixer fitted with a double-paddle impeller and an in-situ pH probe (Figure 2). The CSTR was initially charged with a divalent cation-rich brine having a pH of 7.4 and containing (in mg/L): Na, 11,800; Mg, 4,810; Ca, 61,900; Sr, 4,350; and Cl, 132,100. Alkalization was performed by the stepwise addition of a 50 wt% sodium hydroxide solution, with pH continuously monitored throughout the process. Samples were collected at intervals corresponding to progressive alkali additions to evaluate the extent of magnesium and calcium precipitation. Upon completion of the magnesium alkalization stage, the slurry was filtered using a dual-pod filtration system equipped with 2 µm filters to remove the precipitated solids. The filtrate was then subjected to further alkalization to target calcium removal. Brine samples from each stage were analyzed for their ionic composition using ion chromatography (IC) and inductively coupled plasma-atomic emission spectroscopy (ICP-AES). Detailed procedures for sample preparation, calibration, and other analytical details are described in our earlier work (Tuhami et al., 2025).

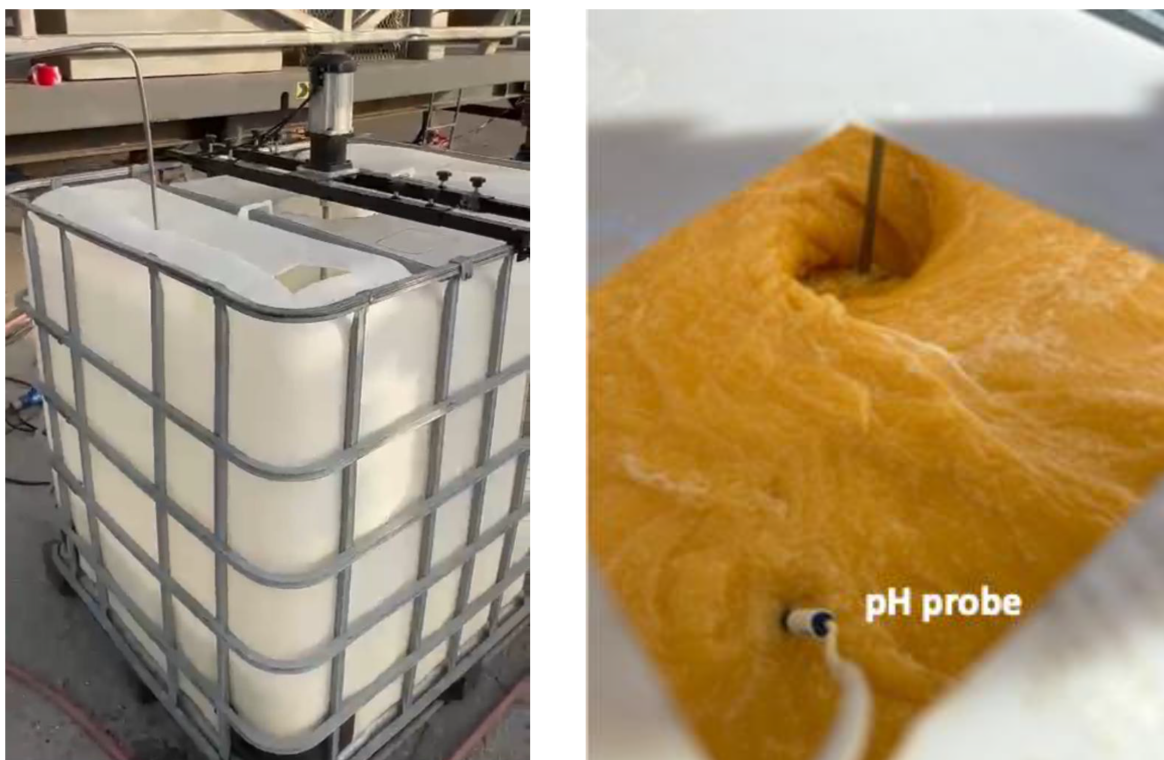


Figure 2—CSTR reactor in the yard test with in-situ pH measurements.

Chemical Thermodynamic Modeling

The equilibrium calculations were performed using the PHREEQC 3 geochemical code (Parkhurst and Appelo, 2013) to examine the system behavior under varying temperatures, pressures, ionic strengths, and calcium hydroxide/water solid-to-liquid-ratio. A Python-based automation framework was developed to systematically generate PHREEQC input scripts, run the simulations, and extract relevant outputs, enabling extensive exploration of the parameter space. This framework utilized the *pitzer.dat* database to capture non-ideal solution behavior through Pitzer activity models for electrolytes (Pitzer 1991), while the gas-phase fugacity was modeled using the Peng–Robinson equation of state (Peng and Robinson, 1976). More details for the used thermodynamic models can be found in our previous work (Alshammari et al 2025).

Results and Discussion

Brine Alkalization Yard Test

The alkalization of brines containing magnesium and calcium leads to the sequential precipitation of magnesium hydroxide (brucite) and calcium hydroxide, governed by their respective solubility equilibria. Magnesium begins to precipitate at a pH of approximately 9, reaching practically complete removal before calcium precipitation initiates at around pH 12. This behavior was systematically investigated in the yard-scale CSTR experiment, with observations aligning closely with the expected chemical thermodynamics. The reactor was thoroughly mixed to eliminate spatial gradients in pH and composition. Figure 3 presents the concentration profiles obtained during the test. Magnesium levels decreased, ultimately falling below the detection limit of 0.1 mg/L, after which the slurry was filtered to remove the precipitated magnesium hydroxide. Following filtration, further addition of sodium hydroxide induced calcium hydroxide precipitation. The initial decline in calcium concentration observed in Figure 3 is attributed to dilution effects from caustic solution addition, while the subsequent steeper decline results from the precipitation reaction. Concurrently, sodium concentrations rose steadily due to the continuous

addition of sodium hydroxide. The pH behavior reflected the sequential saturation of brucite and calcium hydroxide. Initially, the pH decreased slightly and then stabilized around 9, indicative of equilibrium with respect to brucite; continued alkali addition at this stage increased the extent of magnesium precipitation without raising the pH. Once magnesium was effectively depleted, the pH rose and plateaued near 12, consistent with saturation of calcium hydroxide. Samples were collected approximately 10 seconds after each caustic addition, showcasing rapid alkalization kinetics, likely facilitated by the efficient mixing in the reactor. These results demonstrate that pH can serve as a reliable operational parameter to control the sequential removal of magnesium and calcium during the alkalization stage, which makes alkalization an appropriate step in CO₂ mineralization using produced water processes that may involve variation in feed composition.

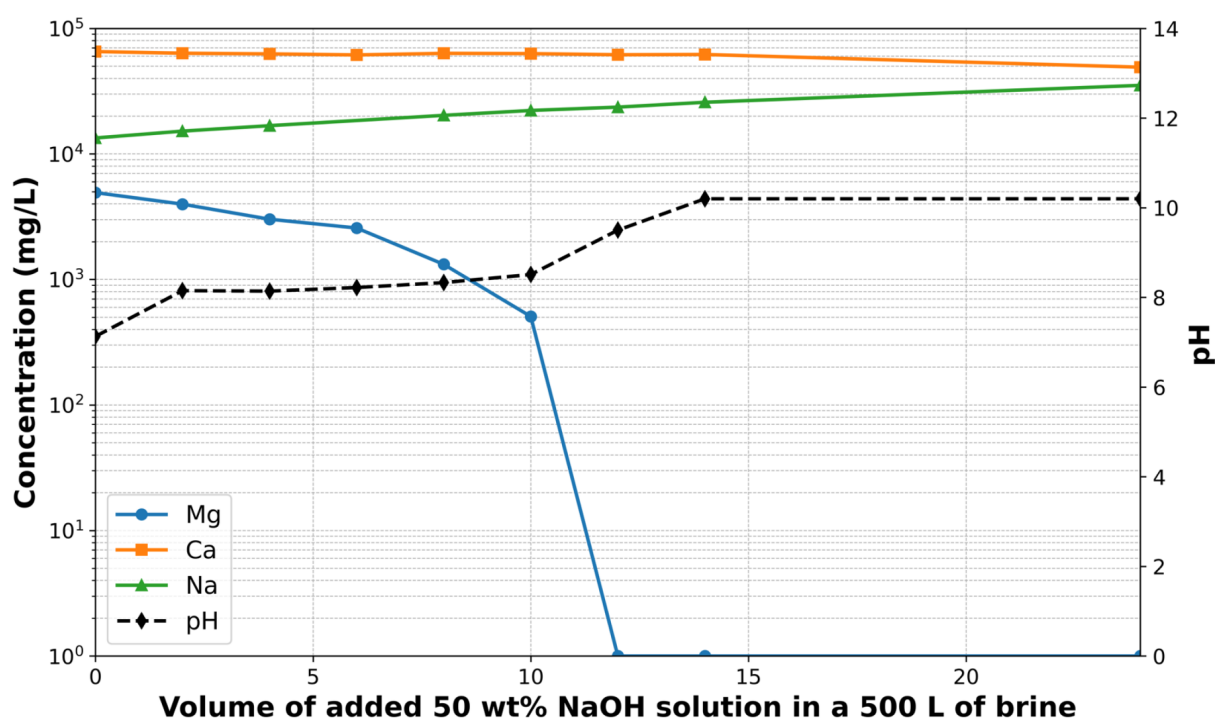


Figure 3—The concentration of sodium, magnesium, and calcium ions in the brine after alkalization with 50 wt% NaOH solution in the yard test.

Mineral Carbonation of Ca(OH)₂

The mineral carbonation of calcium hydroxide involves a sequence of coupled processes, beginning with the mass transfer of carbon dioxide from the gas or supercritical phase into the aqueous phase (R1). This step often represents the rate-limiting stage, governed by factors such as interfacial area between the gas bubbles and liquid, hydrodynamic conditions that dictate convective transport, and the geometric configuration of the reactor.



Subsequently, dissolved carbon dioxide undergoes hydration to form carbonic acid (R2) or reacts directly with hydroxide ions to produce bicarbonate (R3). The relative contribution of these pathways to the depletion of dissolved CO₂ is strongly pH-dependent (Brinkman et al., 1933), with the rate of hydroxylation affected by the chemical potential of hydroxide ions. This is particularly relevant in the carbonation of calcium hydroxide, which acts as a buffer and maintains a consistently high pH of around 12 throughout the process. Such alkaline conditions favor the hydroxylation pathway, thereby enhancing the overall carbonation kinetics.



The buffering capacity imparted by suspended calcium hydroxide arises from its equilibrium with dissolved calcium and hydroxide ions (R4). As carbon dioxide dissolves and forms carbonic acid and bicarbonate, hydroxide ions released from the $\text{Ca}(\text{OH})_2$ equilibrium (R4) facilitate deprotonation of carbonic acid (R5) and bicarbonate (R6), driving the formation of carbonate ions. These carbonate ions subsequently react with calcium ions to precipitate calcium carbonate (R7). Continued carbonation lowers the pH and reduces carbonate activity, eventually leading to partial dissolution of calcium carbonate (R8).



Figure 4 illustrates an example of calcium hydroxide mineral carbonation under mass transfer-limited conditions. In this system, the amount of carbon dioxide in the gas phase was incrementally increased until a fugacity of 1 was achieved. The initial composition comprised 0.05 moles of calcium hydroxide (portlandite) suspended in 1 kg of water, partially dissociating into calcium and hydroxide ions in equilibrium. As CO_2 was introduced, the concentration of calcium ions remained stable while the amount of solid $\text{Ca}(\text{OH})_2$ progressively decreased, eventually reaching complete depletion. Beyond this point, calcium ion concentrations began to decline. Throughout this initial stage, the pH remained constant at approximately 12, sustained by the buffering action of $\text{Ca}(\text{OH})_2$. Concurrently, calcium carbonate precipitated, reaching a maximum before subsequently decreasing due to dissolution driven by the drop in pH and carbonate ion activity. These observations underscore the importance of terminating the carbonation process once the pH begins to fall, to mitigate calcium carbonate dissolution. This consideration is particularly critical at low solid-to-liquid ratios, where even modest dissolution can substantially diminish the final yield of precipitated calcium carbonate.

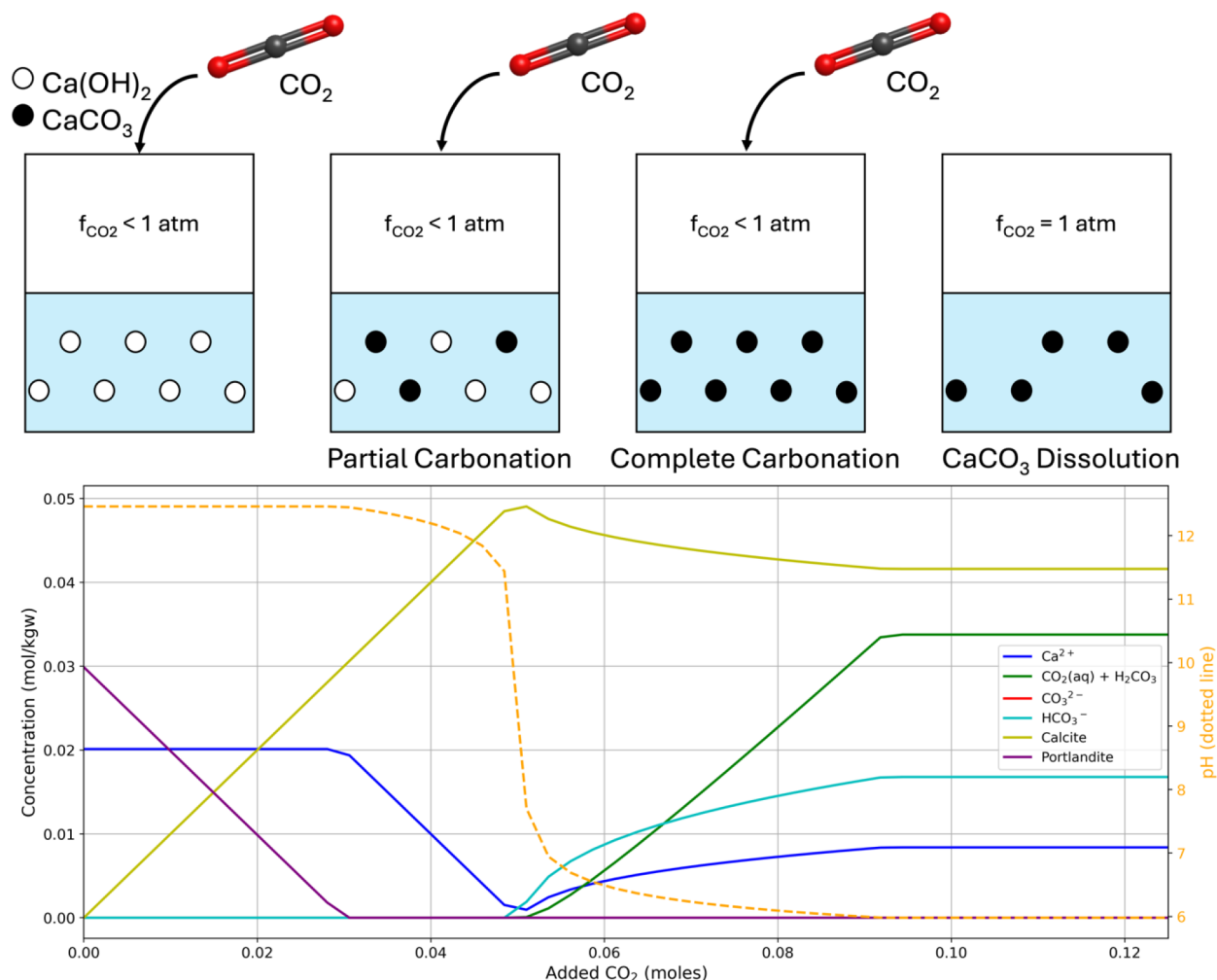


Figure 4—Aqueous speciation during the carbonation of 0.05 mol of calcium hydroxide (portlandite) suspended in 1 kg of water for mass transfer limited systems.

Effect of Temperature, Pressure and Solid-to-Liquid Ratio on Mineral Carbonation Thermodynamics

Figure 5 presents contour maps illustrating the fraction of initial $\text{Ca}(\text{OH})_2$ converted to CaCO_3 as a function of temperature (15–95 °C) and CO_2 pressure (1–300 atm) for varying initial amounts of $\text{Ca}(\text{OH})_2$ (0.01, 0.02, 0.05, 0.1, and 0.2 moles in 1 kg of water). The CO_2 mineralization yield is defined as the number of moles of CaCO_3 precipitated at equilibrium divided by the initial moles of $\text{Ca}(\text{OH})_2$. An increase in temperature was found to enhance the mineralization yield, whereas increasing CO_2 pressure exhibited the opposite effect. Additionally, lower initial concentrations of $\text{Ca}(\text{OH})_2$ in 1 kg of water resulted in reduced conversion to CaCO_3 . For instance, at an initial $\text{Ca}(\text{OH})_2$ loading of 0.01 mol, the CaCO_3 yield was nearly zero across most temperature and pressure conditions (Figure 5a). The reason behind the low yield for low initial modalities is that amount of calcium carbonate per unit volume of water is very low and the extent of its dissolution (R8) will be comparatively high. Therefore, although saturated calcium hydroxide solutions facilitate faster CO_2 absorption compared to suspensions and higher amounts of suspended $\text{Ca}(\text{OH})_2$ can reduce the absorption rate (Han et al., 2011), the design of mineral carbonation processes is ultimately constrained at low solid-to-liquid ratios due to extensive mineral dissolution, which limits overall carbonation efficiency.

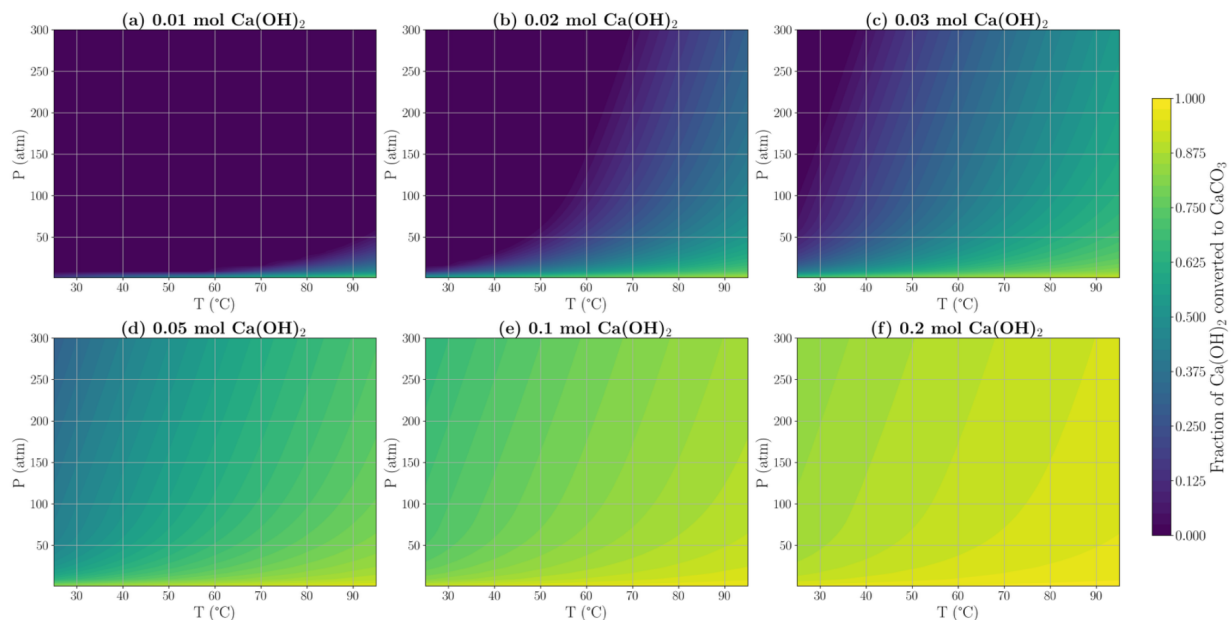


Figure 5—Contour maps for the molar fraction of Ca(OH)_2 converted to CaCO_3 in an initial system of 1 kg of pure water + x moles of Ca(OH)_2 , which gets carbonated with a gas phase of pure CO_2 at different pressures (1 to 300 atm) and temperatures (15 to 95 °C).

The proposed process design, which integrates sequential alkalization with subsequent mineral carbonation, offers several notable advantages. These include the recovery of valuable minerals such as magnesium hydroxide and calcium carbonate, effective carbon dioxide abatement, and a reduction in brine hardness which helps decreasing the propensity for inorganic scale formation and mitigating potential formation damage upon reinjection. When calcium hydroxide is employed for CO_2 mineralization, the overall sequestration capacity is primarily governed by the calcium concentration in the brine and the volumetric flow rate. Assuming a mineralization efficiency of 80%, the sequestration potential was estimated for calcium-rich produced water streams. The results indicate that, for produced water production rates ranging from 100,000 to 1,000,000 barrels per day with a calcium concentration between 2,000 and 20,000, the CO_2 sequestration capacity would lie between approximately 100,000 and 1,000,000 tonnes per annum (Figure 6).

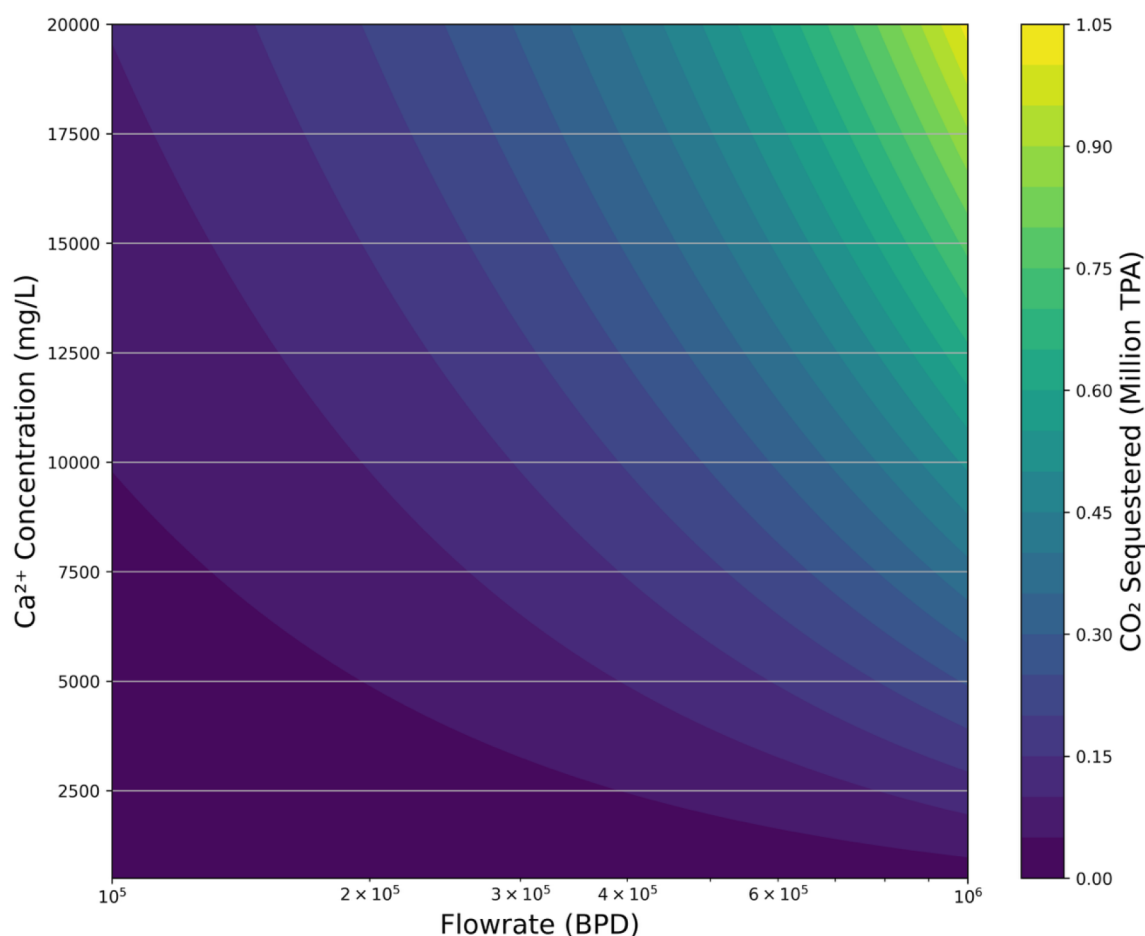


Figure 6—Carbon dioxide sequestration potential as a function of calcium concentration and brine flow rate in barrels per day (BPD).

Conclusion

This study demonstrates the technical viability of utilizing divalent cation-rich produced water for CO₂ sequestration through sequential alkalization and mineral carbonation. Yard-scale experiments conducted in a 600 L continuously stirred tank reactor confirmed the selective precipitation of magnesium and calcium hydroxides under controlled pH regimes, consistent with the thermodynamic predictions. Comprehensive analytical characterization verified the near-complete removal of magnesium at pH 8–9, followed by calcium hydroxide formation above pH 10. Complementary chemical thermodynamic modeling, employing Pitzer equations for electrolyte solutions and the Peng–Robinson equation of state for the gas phase, elucidated the influence of key process variables. Elevated temperatures and reduced pressures were shown to a higher thermodynamic yield for the carbonation of extracted calcium hydroxide to calcium carbonate. Furthermore, stepwise CO₂ addition experiments underscored the importance of carefully managing pH to maximize CaCO₃ precipitation, emphasizing that carbonation progresses at a stable pH value until a certain point marking the consumption of calcium hydroxide, beyond which acidification leads to partial dissolution. This work advances the field by bridging chemical thermodynamic modeling and yard-scale implementation with actual produced brines, providing essential insights for the design of scalable, efficient CO₂ mineralization processes that address operational challenges related to kinetics, feed variability, and optimal process conditions.

References

- Alshammari, S., Abdel-Azeim, S., Al-Yaseri, A. et al 2024a. The Influence of CH₄ and CO₂ on the Interfacial Tension of H₂-Brine, Water-H₂-Rock Wettability, and Their Implications on Geological Hydrogen Storage. *Energy & Fuels* **38** (16): 15834–15847. <https://doi.org/10.1021/acs.energyfuels.4c02234>.
- Alshammari, S., Sangaru, S., Saleem, H. et al 2024b. Lithium Extraction from Produced Water Using Lithium Aluminum Layered Double Hydroxide Chloride Sorbent. Presented at ADIPEC, Abu Dhabi, UAE, 4 November. SPE-222226-MS. <https://doi.org/10.2118/222226-MS>.
- Alshammari, S., Yousif, Z., Alyousef, M. et al 2025. The Extent of Reactive Solubility Trapping During CO₂ Storage in Geological Carbonate Formations. Presented at SPE Advances in Integrated Reservoir Modelling and Field Development Conference and Exhibition, Abu Dhabi, UAE, June 02 2025, SPE-225350. <https://doi.org/10.2118/225350-MS>.
- Alyousef, M., Alshammari, S., and Al-Yaseri, A. 2025. Synergy of CO Mineralization in Produced Water with Enhanced Oil Recovery: An Experimental Study. *Fuel* **382**: 133694. <https://doi.org/10.1016/j.fuel.2024.133694>.
- Bai, M., Sun, J., Song, K. et al 2015. Well completion and integrity evaluation for CO₂ injection wells. *Renewable and Sustainable Energy Reviews* **45**: 556–564. <https://doi.org/10.1016/j.rser.2015.02.022>.
- Bai, M., Zhang, Z., and Fu, X. 2016. A review on well integrity issues for CO₂ geological storage and enhanced gas recovery. *Renewable and Sustainable Energy Reviews* **59**: 920–926. <https://doi.org/10.1016/j.rser.2016.01.043>.
- Bang, J., Chae, S., Lee, S. et al 2019. Sequential carbonate mineralization of desalination brine for CO₂ emission reduction. *Journal of CO₂ Utilization* **33**: 427–433. <https://doi.org/10.1016/j.jcou.2019.07.020>.
- Bang, J., Chae, S., Song, K. et al 2021. Optimizing experimental parameters in sequential CO₂ mineralization using seawater desalination brine. *Desalination* **519**: 115309. <https://doi.org/10.1016/j.desal.2021.115309>.
- Bashir, A., Ali, M., Patil, S. et al 2024. Comprehensive Review of CO₂ Geological Storage: Exploring Principles, Mechanisms, and Prospects. *Earth-Science Reviews* **249**: 104672. <https://doi.org/10.1016/j.earscirev.2023.104672>.
- Brinkman, R., Margaria, R., and Roughton, F. 1933. The kinetics of the carbon dioxide-carbonic acid reaction. *Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character* **232**, no. 707-720 (1933): 65–97. <https://doi.org/10.1098/rsta.1934.0003>.
- Chen, T., Zhao, Y., Zhao, Y. et al 2021. Competitive Ion Migration and Process Optimization of Carbon Sequestration and Seawater Decalcification in a Bipolar Electrodialysis Process. *ACS Sustainable Chemistry & Engineering* **9** (25): 8372–8382. <https://doi.org/10.1021/acssuschemeng.0c09232>.
- Collins, A. 1975 Geochemistry of oilfield waters. Vol. 1. Elsevier.
- Doney, S., Fabry, V., Feely, R. et al 2009. Ocean Acidification: The Other CO₂ Problem. *Annual Review of Marine Science* **1** (1): 169–192. <https://doi.org/10.1146/annurev.marine.010908.163834>.
- Druckemiller, M. and Maroto-Valer, M. 2005. Carbon sequestration using brine of adjusted pH to form mineral carbonates. *Fuel Processing Technology* **86**: 1599–1614. <https://doi.org/10.1016/j.fuproc.2005.01.007>.
- Eikeland, E., Blichfeld, A., Tyrsted, C. et al 2015. Optimized Carbonation of Magnesium Silicate Mineral for CO₂ Storage. *ACS Applied Materials & Interfaces* **7** (9): 5258–5264. <https://doi.org/10.1021/am508432w>.
- Han, S., Yoo, M., Kim, D. et al 2011. Carbon Dioxide Capture Using Calcium Hydroxide Aqueous Solution as the Absorbent. *Energy & Fuels* **25** (8): 3825–3834. <https://doi.org/10.1021/ef200415p>.
- Huntzinger, D., Gierke, J., Kawatra, S. et al 2009. Carbon Dioxide Sequestration in Cement Kiln Dust through Mineral Carbonation. *Environmental Science & Technology* **43** (6): 1986–1992. <https://doi.org/10.1021/es802910z>.
- IPCC. 2023. Climate Change 2023: Synthesis Report. *Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Geneva, Switzerland.
- Kelly, K., Silcox, G., Sarofim, A. et al 2011. An evaluation of ex situ, industrial-scale, aqueous CO₂ mineralization. *International Journal of Greenhouse Gas Control* **5** (6): 1587–1595. <https://doi.org/10.1016/j.ijggc.2011.09.005>.
- Lee, S., Chae, J., and Jung, S. 2024. Enhanced CO₂ Removal Through the Electrolysis of Concentrated Seawater and Accelerated Mineral Carbonation. *KSCE Journal of Civil Engineering* **28** (8): 3109–3119. <https://doi.org/10.1007/s12205-024-1819-4>.
- Ma, J., Li, L., Wang, H. et al 2016. Geophysical monitoring technology for CO₂ sequestration. *Applied Geophysics* **13** (2): 288–306. <https://doi.org/10.1007/s11770-016-0553-1>.
- Montes-Hernandez, G., Pérez-López, R., Renard, F. et al 2009. Mineral sequestration of CO₂ by aqueous carbonation of coal combustion fly-ash. *Journal of Hazardous Materials* **161**: 1347–1354. <https://doi.org/10.1016/j.jhazmat.2008.04.104>.
- Park, A., Jadhav, R., and Fan, L. 2003. CO₂ Mineral Sequestration: Chemically Enhanced Aqueous Carbonation of Serpentine. *The Canadian Journal of Chemical Engineering* **81**: 885–890. <https://doi.org/10.1002/cjce.5450810373>.
- Parkhurst, D.L., and Appelo, C.A.J. 2013. Description of Input and Examples for PHREEQC Version 3 - A Computer Program for Speciation, Batch-Reaction, One-Dimensional Transport, and Inverse Geochemical Calculations. *US Geological Survey Techniques and Methods* 6-A43.

- Peng, D.Y., and Robinson, D.B. 1976. A New Two-Constant Equation of State. *Industrial & Engineering Chemistry Fundamentals* **15** (1): 59–64. <https://doi.org/10.1021/i160057a011>.
- Pitzer, K.S. 1991. Activity Coefficients in Electrolyte Solutions. 2nd edition. Boca Raton, FL: CRC Press.
- Pogge von Strandmann, P., Burton, K., Snæbjörnsdóttir, S. et al 2019. Rapid CO₂ mineralisation into calcite at the CarbFix storage site quantified using calcium isotopes. *Nature Communications* **10** (1): 1983. <https://doi.org/10.1038/s41467-019-10003-8>.
- Santos, R., Van Bouwel, J., Vandeveld, E. et al 2013. Accelerated mineral carbonation of stainless steel slags for CO₂ storage and waste valorization: Effect of process parameters on geochemical properties. *International Journal of Greenhouse Gas Control* **17**: 32–45. <https://doi.org/10.1016/j.ijggc.2013.04.004>.
- Soong, Y., Fauth, D., Howard, B. et al 2006. CO₂ sequestration with brine solution and fly ashes. *Energy Conversion and Management* **47**: 1676–1685. <https://doi.org/10.1016/j.enconman.2005.10.021>.
- Soong, Y., Goodman, A., McCarthy-Jones, J. et al 2004. Experimental and simulation studies on mineral trapping of CO₂ with brine. *Energy Conversion and Management* **45**: 1845–1859. <https://doi.org/10.1016/j.enconman.2003.09.029>.
- Suleman Kali, R., Fan, D., Hazel, N. et al 2022. Techno-socio-economic analysis of geological carbon sequestration opportunities. *Environmental Science: Advances* **1** (2): 138–155. <https://doi.org/10.1039/d1va00036e>.
- Tuhami, M., Al-Saleem, H., Alshammari, S. et al 2025. Coupling Well Log Data with Chemical Thermodynamic Modeling for Predicting Mineral Precipitation Along the Wellbore Depth Profile. Presented at The SPE Advances in Integrated Reservoir Modelling and Field Development Conference and Exhibition, Abu Dhabi, UAE, June 2025, SPE-225407-MS. <https://doi.org/10.2118/225407-MS>.
- Wang, W., Hu, M., Zheng, Y. et al 2011. CO₂ Fixation in Ca²⁺-/Mg²⁺-Rich Aqueous Solutions through Enhanced Carbonate Precipitation. *Industrial & Engineering Chemistry Research* **50** (13): 8333–8339. <https://doi.org/10.1021/ie1025419>.
- Wu, J., Sheen, J., Chen, S. et al 2001. Feasibility of CO₂ Fixation via Artificial Rock Weathering. *Industrial & Engineering Chemistry Research* **40** (18): 3902–3905. <https://doi.org/10.1021/ie010222l>.
- Zevenhoven, R., Wiklund, A., Fagerlund, J. et al 2010. Carbonation of calcium-containing mineral and industrial by-products. *Frontiers of Chemical Engineering in China* **4** (2): 110–119. <https://doi.org/10.1007/s11705-009-0238-x>.
- Zhang, N., Chai, Y., Santos, R. et al 2020. Advances in process development of aqueous CO₂ mineralisation towards scalability. *Journal of Environmental Chemical Engineering* **8** (6): 104453. <https://doi.org/10.1016/j.jece.2020.104453>.
- Zhu, B., Wilson, S., Raudsepp, M. et al 2022. Accelerating mineral carbonation in hydraulic fracturing flowback and produced water using CO₂-rich gas. *Applied Geochemistry* **143**: 105380. <https://doi.org/10.1016/j.apgeochem.2022.105380>.